

3. Pension Funds

The largest pension funds are managed at the sovereign level. In fact, national pension funds can be considered SWFs, although the line is blurry. Australia's and New Zealand's SWFs are explicitly designed to meet future national pension liabilities, and they embrace the SWF moniker. On the other hand, Canada Pension Plan takes great pains to explain why it should not be called a SWF.¹⁸

There are four types of pension savings:

1. National pension plans like Social Security;
2. Private *defined benefit plans*, where future certain, or pre-determined, benefits are promised to beneficiaries by companies;
3. Private *defined contribution plans*, where payments into the plan are predefined, but the future benefits are not fixed; and
4. Funds privately managed by individuals—in the United States, these include IRA, Keogh, and 401(k) plans, which have advantageous tax treatments but strict contribution limits (see chapter 12).

Economists often refer to *pillars* or *tiers* of support when talking about the design of pension systems.¹⁹ Social Security is a first pillar and is designed to provide minimal support for most citizens. The second pillar, consisting of work-related pensions, includes (2) and (3), which capture the bulk of retirement savings in the United States. The third pillar is private voluntary savings, which is (4), but it also includes non-tax advantaged savings used in retirement. And finally there is now a fourth pillar: if retirement savings are inadequate, (great-) grandma may have no choice but to go back to work.

3.1. DEFINED BENEFIT VERSUS DEFINED CONTRIBUTION

In defined benefit plans, the employer pays a retirement benefit based on worker age, years of employment, and current and past wages. Hence, the benefit is “defined.” A typical payout at retirement is something like “benefit factor $\times$ highest average pay over a three-year period $\times$ number of years of service,” where the benefit factor might increase with age. For example, the benefit factor is equal to 2.0% for a teacher retiring at age fifty-five under the California Public Employees’ Retirement System (CalPERS), the largest pension plan in the United States. The benefit factor increases to 2.5% for a teacher sixty-three or older. Since the employee is assured a predictable amount at retirement, the employer (mostly) bears the investment risk.

¹⁸ See www.cppib.ca/files/PDF/SWFBkgr_Dec10_2007.pdf.

¹⁹ This terminology comes from the influential World Bank 1994 report, *Averting the Old Age Crisis*.